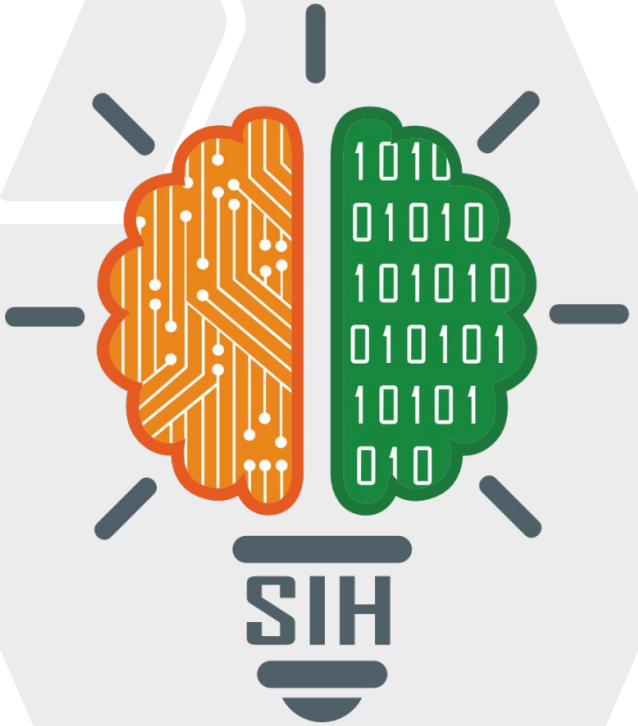
SMART INDIA HACKATHON 202

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NidhiKavach — Security Deposit & FDR Lifecycle Portal

- **Problem Statement ID–** SIH26265
- **Problem Statement Title-** Streamline of Security Deposit / FDR Portal
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-** 198755
- **Team Name-** AIT_G_SPARK



nidhikavach

[every deposit, accounted for]

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NIDHIKAVACH — SMART FDR PORTAL

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■ proposed solution

nidhikavach

[security deposit & fdr portal]

every deposit, released right

[what it is]

A secure AICTE portal where institutions submit each FDR once, officers verify it against the bank's record, and the system watches every deadline until an **authorised, eSigned release**. Covers all 10 capabilities the PS asks for.

[usp]

Three-source truth: every field is checked three ways (declared, scanned, bank-confirmed) and rolled into one trust score. Maker ≠ checker approvals and a **hash-chained audit trail** make fraud visible, not just recorded.

■ the problem

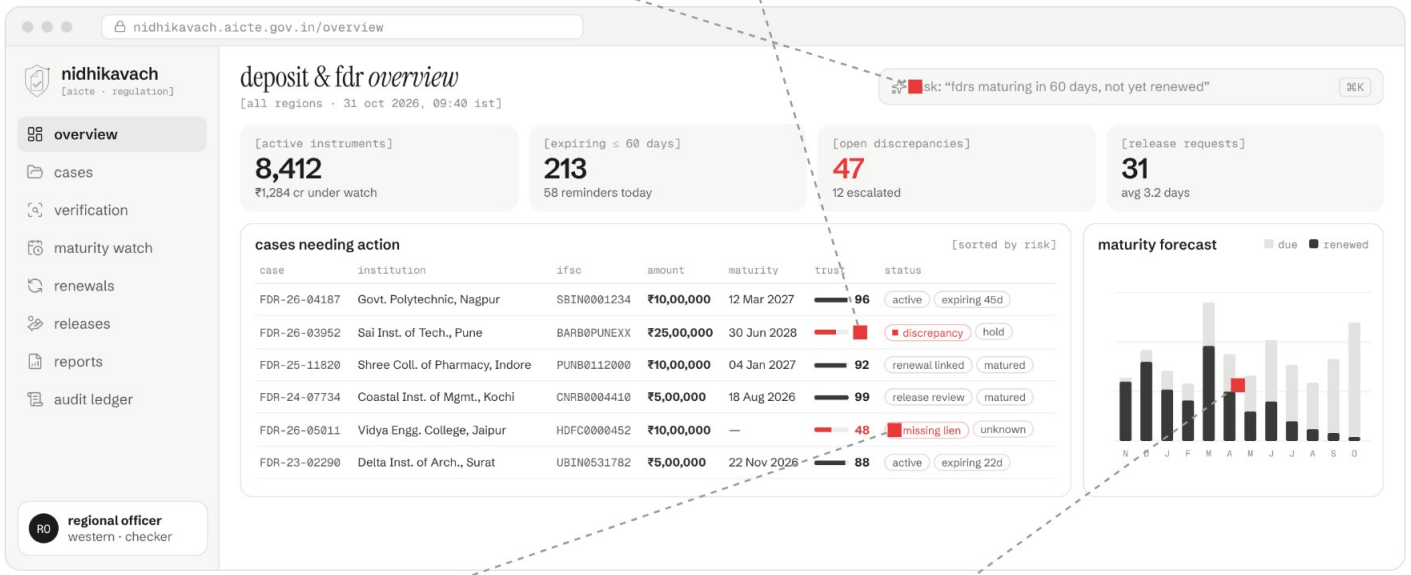
₹116.84 cr
fake fdrs at one civic body
[chandigarh mc · 2026]

₹35 lakh
per programme, held 10 yrs
[aicte aph 2024-27]

3 desks
to clear one release
[aph cl. 6.46]

[ask in plain language — "fdrs maturing in 60 days"]

[trust score from 3 sources: declared · scan · bank]



[auto-flag when the declared record ≠ bank record]

[t-90 → t-7 maturity reminders, auto-escalated]

[github repo] github.com/harshwardhan-kp/nidhikavach

[live demo] nidhikavach.vercel.app

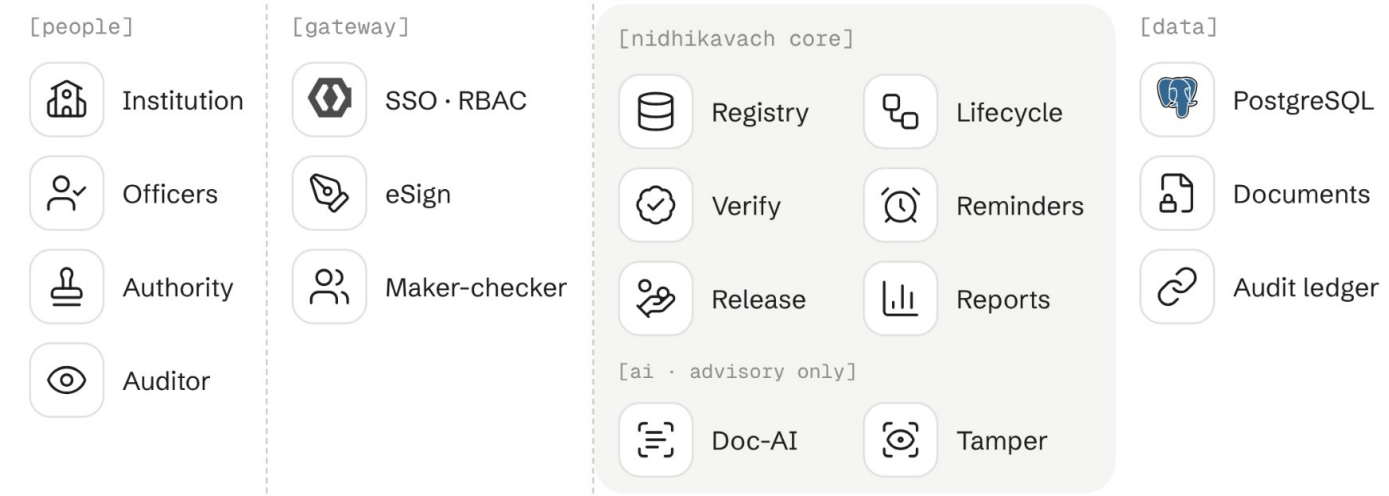
[demo video] drive.google.com/drive/folders/1j3dU2e2X0FqaFsvp0iVqWVg2KX_jNwlF

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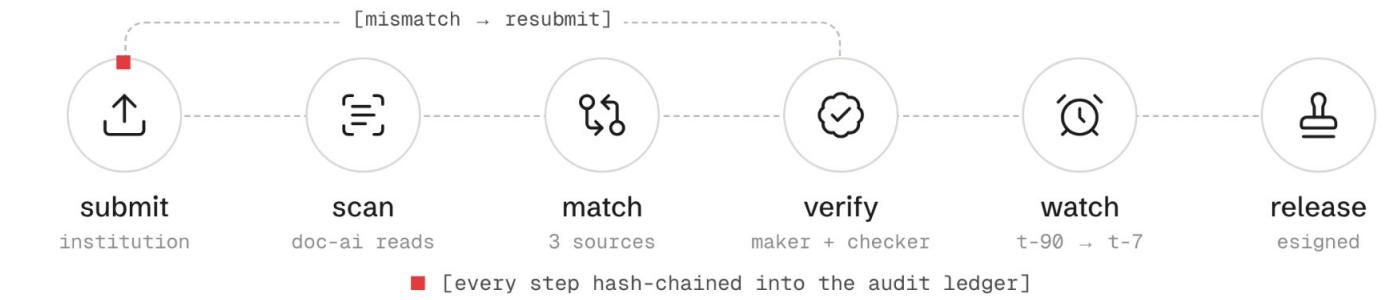
TECHNICAL APPROACH

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methodology — architecture



process — one fdr, end to end



technologies to be used

Next.js · React · TypeScript · Tailwind CSS · shadcn/ui · nest · node · Temporal · FastAPI · python · PostgreSQL · redis · MinIO · OpenSearch · PyTorch · OpenCV · PaddleOCR · Hugging Face · Ollama · Keycloak · Vault · NGINX · docker · kubernetes · Github Actions · GitHub · Prometheus · Grafana · OpenTelemetry · OpenAPI · pytest · Playwright

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FEASIBILITY AND VIABILITY

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analysis of feasibility

[high]

technical
Proven open-source stack

[high]

operational
Fits existing AICTE roles

[high]

economic
₹0 licences · NIC cloud

[high]

legal
IT Act · DPDP 2023 · CERT-In

challenges & risks → strategies

[risk]		[strategy]
no api for every bank	→	AA consent + manual maker-checker
messy legacy records	→	Guided import — unknowns never guessed
forged fdr scans	→	Tamper scan + bank confirmation
insider self-approval	→	Maker ≠ checker, by identity
aph rules change	→	Versioned rule engine
ai mistakes	→	Advisory only — humans decide

path to deployment

sih prototype
[36 h]

regional pilot
[3 months]

national rollout
[6-9 months]

bank & e-bg links
[ongoing]

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IMPACT AND BENEFITS

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potential impact – pilot targets

days → hours	0	100%	1 record
[verification turnaround]	[maturities missed]	[cases with an owner & next step]	[per fdr, full history]

[measured against a manual baseline on the same sample cases, as the problem statement asks]

who gains

Regulation Bureau
[live, risk-ranked view]

Institutions
[upload once, get reminded]

Banks
[fewer verification letters]

Students & public
[deposits kept safe]

benefits

social
trust in regulator

economic
less fraud, delay

environmental
paperless fdrs

governance
audit-ready

how we compare

	[files + email + excel]	[generic e-office / dms]	nidhikavach
Single record per FDR	○	◐	●
Declared vs bank-confirmed	○	○	●
Automatic maturity alerts	○	◐	●
Renewals linked to history	○	◐	●
Maker-checker release + eSign	◐	◐	●
Tamper-evident audit trail	○	◐	●
Forged-scan detection	○	○	●

● built in ◐ partial / manual ○ not available

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RESEARCH AND REFERENCES

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details & links of references

problem & regulation

SIH26265 · AICTE Regulation Bureau
[sih.gov.in]

AICTE Approval Process Handbook 2024-27 — Cl. 1.12, Table 1.4, Cl. 6.46
[annauniv.edu/cai/APH2024-25.pdf]

342 AICTE institutions closed, 2024-27 (Lok Sabha)
[news.careers360.com · aug 2026]

AICTE "Light but Tight" compliance reform
[aicte.gov.in · oct 2021]

evidence: deposit fraud

₹116.84 Cr fake FDRs — Chandigarh MC
[outlookindia.com · mar 2026]

₹145 Cr FDR mismatch — Panchkula MC
[outlookbusiness.com · mar 2026]

AICTE joint FDRs encashed without consent (CBI)
[deccanherald.com · dec 2009]

₹13.24 Cr lost to fake bank guarantees — CAG 1/2021
[cag.gov.in]

verification rails

RBI Account Aggregator Master Direction — FDs in scope
[rbi.org.in]

ReBIT Term Deposit schema v1.2 · Sahamati AA stats
[rebit.org.in · sahamati.org.in]

NeSL e-BG · e-FDR with API-only lien release
[mondaq.com · bankofbaroda.bank.in]

IFSC dataset from RBI NEFT/RTGS lists
[github.com/razorpay/ifsc]

standards & build

DPDP Act 2023 · IT Act 2000 (eSign) · CERT-In
[meity.gov.in · cert-in.org.in]

GIGW 3.0 accessibility guidelines
[guidelines.india.gov.in]

e-Office: 93.78% of central files digital (DARPG)
[devdiscourse.com · mar 2026]

Temporal · Keycloak · PaddleOCR docs
[temporal.io · keycloak.org · github.com]

research work behind this idea

ps + aph	10 / 10	6 cases	5 roles	5 lanes
[read clause by clause]	[capabilities mapped]	[of fdr fraud studied]	[permission matrix]	[status model]

nidhikavach [every deposit, accounted for]